

CSA1703 – Artificial Intelligence

Name:- B. Deepthi reddy

Reg No:- 192312288

Slot – A

3 .WATER JUG PROBLEM

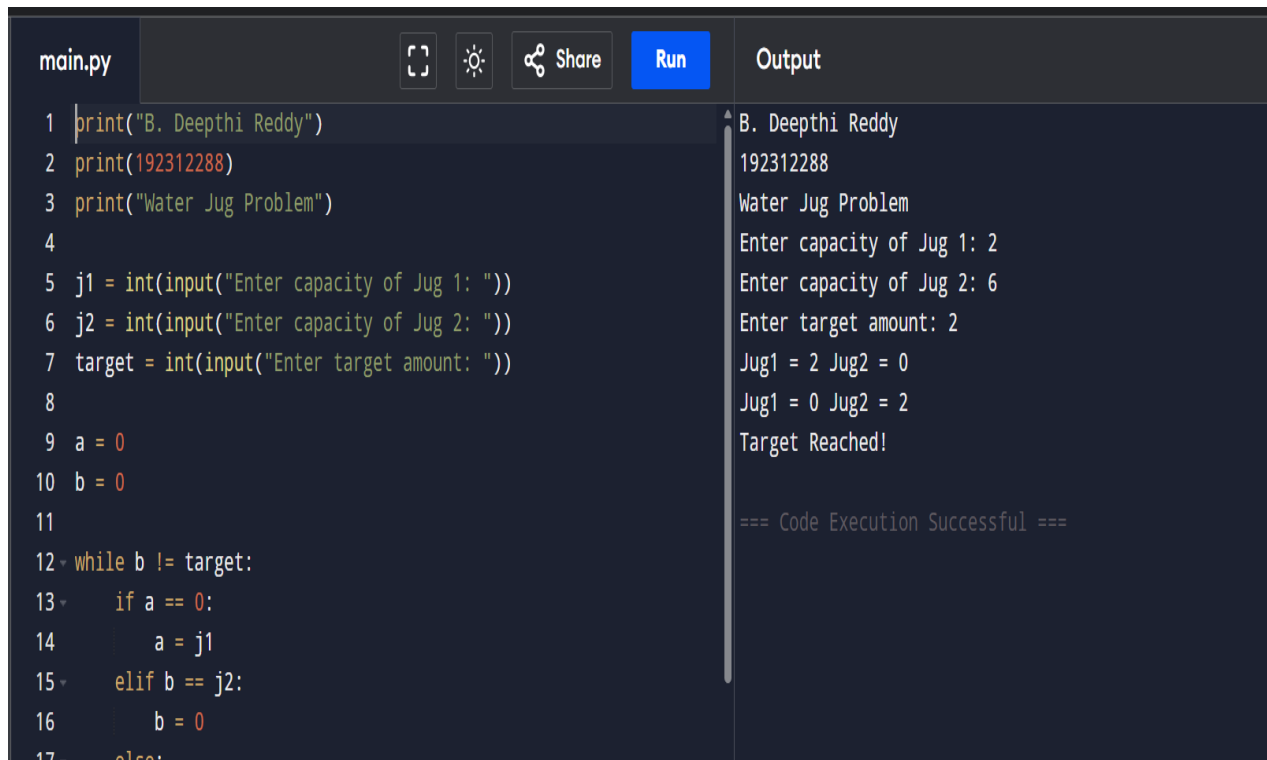
Code:

```
print("B. Deepthi Reddy")
print(192312288)
print("Water Jug Problem")
j1 = int(input("Enter capacity of Jug 1: "))
j2 = int(input("Enter capacity of Jug 2: "))
target = int(input("Enter target amount: "))
a = 0
b = 0
while b != target:
    if a == 0:
        a = j1
    elif b == j2:
        b = 0
    else:
        t = min(a, j2 - b)
        a -= t
        b += t
```

```
print("Jug1 =", a, "Jug2 =", b)
```

```
print("Target Reached!")
```

Output:



The screenshot shows a code editor with a file named 'main.py'. The code is a Python script for the Water Jug Problem. It includes print statements for the name, ID, and problem title, followed by input prompts for jug capacities and target amount. It then uses a while loop to simulate the water transfer process between two jugs. The output window on the right shows the execution results, including the user input and the final state of the jugs.

```
main.py  [Full Screen] [Theme] [Share] [Run] Output
```

```
1 print("B. Deepthi Reddy")
2 print(192312288)
3 print("Water Jug Problem")
4
5 j1 = int(input("Enter capacity of Jug 1: "))
6 j2 = int(input("Enter capacity of Jug 2: "))
7 target = int(input("Enter target amount: "))
8
9 a = 0
10 b = 0
11
12 while b != target:
13     if a == 0:
14         a = j1
15     elif b == j2:
16         b = 0
17     else:
```

Output:

```
B. Deepthi Reddy
192312288
Water Jug Problem
Enter capacity of Jug 1: 2
Enter capacity of Jug 2: 6
Enter target amount: 2
Jug1 = 2 Jug2 = 0
Jug1 = 0 Jug2 = 2
Target Reached!

=== Code Execution Successful ===
```